

Block One Synthesis. The Hepatologist's Toolkit.

Imagine a single clinic morning. A six-week-old girl arrives with jaundice and stools her parents call pale yellow. A well four-year-old boy is referred because his aspartate aminotransferase, called AST, came back high on blood tests taken before a minor operation. A twelve-year-old girl with newly diagnosed autoimmune hepatitis has a liver stiffness reading that the report calls consistent with cirrhosis. A nine-month-old boy after a Kasai operation needs a decision about a liver biopsy, and his international normalised ratio, called INR, is one point nine. And a three-month-old with intense itch has a genetic report that names one disease-causing variant and one variant of uncertain significance.

Every one of these children needs a test, or a set of tests, from the toolkit we have built over the last thirteen lessons. None of them needs every test. This chapter brings the block together. It asks one question: which tool answers which question, and in what order? The main idea is simple to say and hard to practise. Every test in hepatology measures one biological process. It answers one question well, it answers others badly, and each has a typical way of being fooled. The expert is not the person who knows the longest list of tests. The expert is the person who names the question before ordering, knows what would change if the answer came back one way or the other, and knows what could make the result lie.

We will do four things. First, we will go through the toolkit one family at a time and name the blind spot of each tool. Second, we will compare the anchor guidelines of the block and see where they agree and where they do not. Third, we will work through five difficult cases. Fourth, we will close with twenty pearls, a list of what we still do not know, and a short set of questions to test yourself.

A word on the documents. The main paediatric guideline for the jaundiced infant is the 2017 joint guideline of the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, called NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, called ESPGHAN, written by Fawaz and colleagues. For liver biopsy in children the anchor is the 2015 position paper of the ESPGHAN Hepatology Committee, written by Dezsofi and colleagues. For elastography there are two ultrasound documents: the 2017 update from the European Federation of Societies for Ultrasound in Medicine and Biology, called EFSUMB, written by Dietrich and colleagues, and the 2024 paediatric guidance from the World Federation for Ultrasound in Medicine and Biology, called WFUMB, written by Ferraioli and colleagues. Finally, the American Association for the Study of Liver Diseases, called AASLD, published two guidance documents on non-invasive liver disease assessment in 2024, one on blood tests and one on imaging, written by Sterling and colleagues.

Part one. Every test has a blind spot.

Start with the blood. The aminotransferases, alanine aminotransferase or ALT, and AST, leak out of injured hepatocytes. They tell us that liver cells are being damaged, but not how well the liver is working. AST is also found in heart and skeletal muscle and in red cells, so an AST that is out of proportion to the ALT should make us think first of muscle, of haemolysis, and of a harmless curiosity called macro-AST, in which AST is bound to an immunoglobulin and cannot be cleared. The two enzymes also have different half-lives. AST has a half-life of about seventeen hours and ALT about forty-seven hours, so after a single insult AST falls first. Reading the panel as a time series, rather than as a single snapshot, tells you more than any one value.

Gamma-glutamyl transferase, called GGT, and alkaline phosphatase, called ALP, reflect injury to the bile canaliculi and the bile ducts. ALP also comes from bone, so a growing child, especially in puberty, can have a high ALP with a healthy liver. GGT is physiologically high in young infants, which matters when we try to use it in a formula or a score. Neither enzyme tells us where an obstruction is.

Bilirubin must always be split into its fractions. The direct bilirubin, as most laboratories measure it, contains the conjugated bilirubin plus a fraction called delta bilirubin, which is conjugated bilirubin covalently bound to albumin. Delta bilirubin shares the half-life of albumin, about three weeks. That is why jaundice can outlast the disease: the liver has recovered, but the bound bilirubin is still being cleared at the slow pace of albumin. Haemolysis in the sample can also lower the measured direct bilirubin and hide cholestasis.

The INR reflects the vitamin K dependent clotting factors two, seven and ten, together with fibrinogen. In a cholestatic child, vitamin K is poorly absorbed because there is not enough bile in the gut to absorb fat-soluble vitamins. So a prolonged INR in a jaundiced infant is often vitamin K deficiency, not liver failure. And even when the INR does reflect poor synthesis, it does not tell us who will bleed during a procedure, because the liver also makes less of the natural anticoagulants. This is the concept of rebalanced haemostasis.

Now the stiffness tools. Elastography measures how stiff the whole organ is, not how much scar it contains. Anything that makes the liver tense will raise the number: inflammation, cholestasis, venous congestion as in a child after a Fontan operation, and even a recent meal. The 2017 EFSUMB update lists exactly these confounders. The serum scores, such as the AST to platelet ratio index, called APRI, the fibrosis four index, called FIB-4, and the GGT to platelet ratio, called GPR, were all derived in adults with viral hepatitis. They combine a liver enzyme with the platelet count, so they are fooled by anything that lowers platelets, and in infants by the normal surge of GGT. None of them has a validated paediatric cut-off.

The imaging tools each answer a narrow question. The hepatobiliary scan, called HIDA, shows whether bile reaches the gut. When it does, biliary atresia is very unlikely. When it does not, the result is not specific, because prematurity, parenteral nutrition, a shortage of bile ducts, and alpha-1 antitrypsin deficiency can all stop excretion. Magnetic resonance cholangiopancreatography, called MRCP, shows the ducts without an instrument, but in an infant the common duct is tiny and may be collapsed, so a duct that is not seen is not proof of atresia.

Then tissue and code. A liver biopsy shows the pattern of injury, but in a sample that is about one fifty-thousandth of the liver. It is fooled by a small core, by sampling error, and by timing, because some features, such as a shortage of bile ducts or the globules of alpha-1 antitrypsin deficiency, appear only later in infancy. A gene panel looks only at the genes on its list, and its most common output is not a diagnosis but a variant of uncertain significance, or two variants whose relationship to each other is unknown.

Part two. The anchor documents and where they disagree.

Let us take the documents in turn and pick out the line worth remembering from each.

The 2017 NASPGHAN and ESPGHAN guideline on the cholestatic infant has ten numbered recommendations, most of them rated one A or one B, which means strong recommendations based on high or moderate quality evidence. The line to remember is that a direct or conjugated bilirubin above one milligram per decilitre, which is seventeen micromoles per litre, is pathological whatever the total bilirubin, and must be investigated. The guideline asks that formula-fed infants still jaundiced at two weeks be tested, and allows a well breastfed infant to be observed until three weeks. It calls the intraoperative cholangiogram, with histology of the duct remnant, the gold standard for biliary atresia. It calls a properly timed liver biopsy, read by an experienced paediatric pathologist, the most supportive single test, and experienced pathologists reach the correct diagnosis in about ninety to ninety-five percent of cases.

The 2015 ESPGHAN position paper on liver biopsy covers indications, technique and safety. Its practical numbers are a core of about twenty millimetres long and one point eight millimetres wide, at least ten portal tracts or two centimetres of tissue when biliary atresia is the question, correction of clotting or a different route when the INR is above one point five, and observation for at least six hours after the procedure, with bed rest for at least the first hour.

The 2017 EFSUMB update is mostly built on adult data and graded with the Oxford levels of evidence. Its contribution to paediatrics is the list of confounders: food, inflammation, cholestasis and congestion raise stiffness independent of fibrosis. The 2024 WFUMB paediatric guidance is more direct about children. It says that no paediatric cut-off can be recommended, at level of evidence two A and grade B, and yet it offers a working rule: a value of five kilopascals or less rules out significant disease, and fifteen kilopascals or more rules it in, except in biliary atresia, cystic fibrosis liver disease and the Fontan circulation. The AASLD 2024 guidance agrees that there is no preferred method and no threshold for children, and it adds that a single test should not be used on its own to judge whether fibrosis is progressing or regressing.

Now the disagreements. For the cholestatic infant, the threshold itself differs. The North American and European guideline uses one milligram per decilitre of direct or conjugated bilirubin. The British guidance from the National Institute for Health and Care Excellence, called NICE, uses a conjugated bilirubin above twenty-five micromoles per litre, which is about one and a half milligrams per decilitre. The Italian society uses one milligram per decilitre or more than twenty percent of the total when the total is above five. The age at testing also differs a little: the American

Academy of Pediatrics in 2022 allows breastfed infants three to four weeks, and NICE uses fourteen days for term and twenty-one days for preterm infants. These are differences in detail.

The material disagreement is about screening. The 2017 guideline describes stool colour cards and notes that the Taiwanese programme detected pale stools with a sensitivity of about ninety-five percent, but it does not ask for universal screening. The 2025 guidance from the Federation of International Societies of Pediatric Gastroenterology, Hepatology and Nutrition, called FISPGHAN, does recommend universal screening, either with stool colour cards or with an early direct bilirubin. That is a difference that would change practice in most of North America and Europe.

The imaging documents also differ. The 2017 guideline says the HIDA scan should not be used alone to diagnose atresia, but that clear excretion may exclude it. The 2021 European Society of Paediatric Radiology, called ESPR, goes further and says HIDA is not useful for atresia. The 2017 guideline gives MRCP, endoscopic retrograde cholangiopancreatography, called ERCP, and percutaneous cholangiography only a limited role, while the ESPR flowchart uses percutaneous cholangiography or ERCP, with biopsy, in the equivocal case. On the biopsy itself, on early genetic testing, and on the gold standard of the intraoperative cholangiogram, the documents agree.

For tissue and stiffness, the disagreements are sharper. On core size, ESPGHAN asks for twenty millimetres, AASLD asks for at least eleven complete portal tracts for staging, and the British Royal College of Pathologists sets a minimum of ten millimetres and six portal tracts. On clotting, the documents contradict each other. ESPGHAN in 2015 asked for correction of an INR above one point five, or a transjugular or laparoscopic route. The AASLD guidance of 2021 on coagulation in cirrhosis says the INR should not be corrected before procedures, because it does not predict bleeding. The British Society of Gastroenterology, called BSG, sends patients with an INR above one point four to a transvenous route. On observation, ESPGHAN asks for six hours and the BSG for three.

The stiffness and serum score documents split on paediatric fatty liver disease, now called metabolic dysfunction associated steatotic liver disease, or MASLD. ESPGHAN in 2026 says that a transient elastography value below five point five kilopascals excludes significant fibrosis and above ten suggests advanced disease, while AASLD in 2025 says adult thresholds are not validated in children. ESPGHAN in 2026 does not recommend APRI or FIB-4 in paediatric fatty liver disease, while the AASLD blood test guidance of 2024 suggests, without a formal grade, that APRI or FIB-4 may be used to detect advanced fibrosis in children. And WFUMB treats a change of more than ten percent on the same machine as real, while AASLD warns against using a single test to track change.

When we added up all the verdicts from the guideline tables in the first thirteen lessons, one hundred and forty-eight questions in all, the picture was striking. Only forty-one, about twenty-eight percent, were fully aligned. Thirty-nine, about twenty-six percent, differed in detail. Thirty-six, about a quarter, were questions on which only one region or one age group had said anything at all. Twenty-five, about seventeen percent, were material or outright contradictions.

And on seven questions, about five percent, nobody had any data. In other words, agreement was the minority. The commonest reason for a split was not that two groups read the same evidence differently. It was that paediatric evidence was missing, and someone filled the gap by borrowing an adult rule. The practical lesson is to say so out loud: when the paediatric document is silent, name the adult rule you are borrowing and explain why.

Part three. The evidence, test by test, for biliary atresia.

Biliary atresia is the disease that most tests in this block were designed around, so it is worth lining up their specificity. Specificity tells us how often a positive result is truly atresia. The triangular cord sign on ultrasound, a band of echogenic tissue at the porta hepatis, has a specificity of about ninety-seven to one hundred percent. ERCP in expert series has a specificity of about eighty-seven to ninety-four percent. A HIDA scan that shows no excretion has a specificity of only about sixty-eight to seventy-two percent. And MRCP, when it fails to show the duct, has a specificity of only about thirty-six percent.

Sensitivity runs in the opposite direction. The triangular cord sign misses many infants with atresia, with a sensitivity anywhere from twenty-three to eighty-five percent in different series. A HIDA scan rarely misses it: a non-draining scan has a sensitivity of about ninety-nine percent. MRCP is also sensitive, at about ninety-nine percent, but it calls many normal infants abnormal. The rule follows from these numbers. Use the sensitive test to rule out, use the specific test to rule in, and use the cholangiogram to decide.

Four more numbers from the block deserve to be remembered because each one changes what happens to a child. In a study of forty-four well children with an isolated raised AST, seventeen, or about thirty-nine percent, had macro-AST. In infants under three months who had a liver biopsy, the complication rate was about eighteen percent, and seven of the twelve complications came from sedation rather than from the needle. The diagnostic yield of genetic testing for cholestasis was seventy-one percent in a small Japanese cohort selected for a genetic-looking phenotype, but fell to six point eight percent when a large American seventy-seven gene panel was opened to all referred children. And in the Paris cohorts, high-risk varices were present in ninety-six percent of children who had bled, but in only eleven percent of children who never bled. In each case, the number only makes sense when you know the denominator behind it.

Part four. Five difficult cases.

The first case is the six-week-old girl. She has been jaundiced since two weeks of age. Her total bilirubin is seven point eight and her direct bilirubin is two point four milligrams per decilitre. Her GGT is five hundred and twenty units per litre. The ultrasound shows a gallbladder two point two centimetres long without a triangular cord. The HIDA scan shows no excretion at twenty-four hours, and the MRCP does not show the common duct.

Every non-invasive test is equivocal. The direct bilirubin is clearly abnormal, at any total. A normal-looking gallbladder does not exclude atresia, because a normal ultrasound does not rule out the non-syndromic form, and the gallbladder can look normal at six weeks and abnormal at eight.

The HIDA scan is only about seventy percent specific, and the MRCP only thirty-six percent. On the first day we should also exclude the treatable emergencies that can mimic atresia, which are sepsis, galactosaemia, hypothyroidism and hypopituitarism, send the alpha-1 antitrypsin phenotype, and give vitamin K. We should not delay with phenobarbital before a repeat HIDA scan, which costs days and has no randomised evidence of benefit. The decision is a liver biopsy now, read by a paediatric pathologist, followed by a cholangiogram. At six weeks every day counts.

The second case is the four-year-old boy with an AST of two hundred and ten units per litre, a normal ALT, and a normal GGT, confirmed three months later. The referral asks for a liver work-up and a biopsy. But AST without ALT is not liver until proven. In a boy, the first test is creatine kinase, because a muscular dystrophy can present exactly this way. If creatine kinase, lactate dehydrogenase and a haemolysis screen are normal, the next test is polyethylene glycol precipitation, which detects AST bound to immunoglobulin. When more than about eighty-two percent of the AST activity precipitates, macro-AST is likely. In the original paediatric series, macro-AST was still present after six years in six of nine children, and all of them were well. Familial forms are common, and a variant in the AST gene has been found in families, so testing the parents is often the quickest confirmation. The decision is creatine kinase, then precipitation, then the parents. No imaging, no biopsy, and a clear letter that prevents the next referral.

The third case is the twelve-year-old girl with newly diagnosed autoimmune hepatitis. Her ALT is eight hundred and twenty, her bilirubin is three point one, and her immunoglobulin G is twenty-four grams per litre. Her platelets are two hundred and forty thousand, her spleen is normal for her height, and the ultrasound shows no collateral veins. Her transient elastography reading is nineteen point four kilopascals with good quality. The report says consistent with cirrhosis. But inflammation and cholestasis can each push a liver with normal architecture above fifteen kilopascals. The normal platelets and spleen argue against portal hypertension. Autoimmune hepatitis needs a biopsy for diagnosis in any case, and the biopsy, not the probe, gives today's stage. The stiffness should be measured again in biochemical remission, on the same machine with the same probe, and only a change of more than about ten percent should be treated as real. The decision is not to label her cirrhotic.

The fourth case is the nine-month-old boy, weighing seven and a half kilograms, after a Kasai operation at nine weeks that only partly cleared his jaundice. His bilirubin is four point five, his albumin three point two, his INR one point nine, his platelets eighty-five thousand, and his fibrinogen one hundred and sixty milligrams per decilitre. The team asks for a percutaneous biopsy to stage the fibrosis before listing for transplant, and for a screening endoscopy. The first step is parenteral vitamin K, with a repeat INR the next day. If the INR remains prolonged, factor five helps to separate vitamin K deficiency from failing synthesis, because factor five does not depend on vitamin K. Next, we should remember that the INR does not predict bleeding at biopsy, although the ESPGHAN paper still asks for correction or a different route above one point five. The more important question is whether a fibrosis stage would change the decision to list. Often it would not. If a biopsy is still needed, a child under ten kilograms should have the needle tract

plugged, or a transjugular approach, in an expert centre, with six hours of observation. As for the varices, three quarters of first variceal bleeds in biliary atresia happen before two and a half years of age, and the appearance of the varices, not the platelet count, predicts who will bleed. So the decision is vitamin K, fat-soluble vitamin levels, a transplant assessment now, a biopsy only if it changes the plan, and an endoscopy with band ligation if high-risk varices are found.

The fifth case is the three-month-old girl with intense itch, a direct bilirubin of five point two, a GGT of only thirty-five, and an ALT of one hundred and sixty. Her serum bile acids are two hundred and ten micromoles per litre and her fat-soluble vitamins are low. The gene panel reports a frameshift variant in ABCB11, the gene for the bile salt export pump, which is clearly disease-causing, and a missense variant in the same gene of uncertain significance. Their phase is unknown, which means we do not know whether they sit on the same copy of the gene or on opposite copies. Low GGT cholestasis narrows the field to a handful of transporter and tight junction defects and to disorders of bile acid synthesis. The very high serum bile acids argue against a synthesis defect, because in those disorders the serum bile acids are low or normal. One pathogenic variant plus one variant of uncertain significance is the commonest genetic dilemma in this field. The first step is to test both parents, which tells us whether the variants are on opposite copies. The second is to test the tissue: immunostaining for the bile salt export pump on the liver biopsy can show whether the protein is present. If both variants turn out to be truncating, the disease is the severe form sometimes called BSEP three, which is unlikely to respond to the drugs that block the ileal bile acid transporter, and the drug labels for those agents differ by age and by region. The decision is to treat now, with medium chain triglycerides, vitamins and itch control, and to let the genetic work-up guide the choice of drug, never delay it.

Part five. Twenty pearls.

Here are the twenty points of this block that are worth carrying into every clinic.

One. Read the liver panel as a time series. AST, with a half-life of about seventeen hours, falls before ALT, with a half-life of about forty-seven hours.

Two. For an isolated raised AST, check creatine kinase first and then test for macro-AST, which was found in about thirty-nine percent of well children with this pattern.

Three. A direct or conjugated bilirubin above one milligram per decilitre is pathological whatever the total.

Four. Delta bilirubin shares albumin's three week half-life, so jaundice outlasts recovery.

Five. Give parenteral vitamin K before calling any INR a sign of synthetic failure, and use factor five to decide.

Six. The INR measures prognosis, not bleeding. Because haemostasis is rebalanced, a child can form a clot with an INR of two.

Seven. Ask the pathologist to name the zone of injury before the disease, and always give the pathologist the age, the GGT and the question.

Eight. About six portal tracts are enough to make a diagnosis, but about eleven are needed to stage fibrosis, and a one centimetre core under-stages most livers.

Nine. In the young infant, most of the risk of a liver biopsy comes from sedation: complications reached eighteen percent under three months of age.

Ten. Act on a positive triangular cord sign, which is ninety-seven to one hundred percent specific, but ignore a negative one.

Eleven. Excretion on a HIDA scan excludes biliary atresia; non-excretion is only about seventy percent specific.

Twelve. In an infant, seeing the common duct on MRCP excludes atresia, but not seeing it proves nothing.

Thirteen. Normal transient elastography in a child is below about five and a half kilopascals, and sedation, the small probe and liver fat all raise it.

Fourteen. Inflammation, cholestasis and congestion can each push a liver with normal architecture above fifteen kilopascals.

Fifteen. APRI, FIB-4 and the GGT to platelet ratio were born in adult viral hepatitis, and their cut-offs should not be imported into children.

Sixteen. Genetic yield follows the denominator: seventy-one percent in a selected genetic phenotype, under seven percent in open referral.

Seventeen. Send the trio, which means the child and both parents. Re-analyse a negative exome about every three years, as the 2024 guideline of the European Association for the Study of the Liver, called EASL, advises, and never wait for a result to start treatment.

Eighteen. Start energy intake at about one hundred and thirty percent of the requirement for age in a cholestatic child, and measure the triceps skinfold, which finds depletion that weight misses.

Nineteen. Three quarters of first variceal bleeds in biliary atresia happen before two and a half years of age, so screen early.

Twenty. No paediatric non-invasive rule is yet safe enough to cancel a screening endoscopy. The best has a negative predictive value of only about sixty-three percent.

Part six. What we still do not know.

The block also showed us the edges of the evidence. We have no safe INR or platelet threshold for liver biopsy in children, so plasma is still given to treat a number. A prospective paediatric biopsy registry that included viscoelastic tests of clotting would close that gap. We have no

disease-specific and machine-specific stiffness cut-offs for children, and adult values misclassify them; paediatric cohorts with biopsy, analysed by disease and by platform, are needed. We have no fibrosis score derived in children, and the obvious places to build one are biliary atresia, cystic fibrosis and paediatric fatty liver disease. We do not know the normal number of bile ducts per portal tract in the newborn, because the classic ratio below zero point five was set in older children; normal counts on cytokeratin seven staining by gestational age would answer it. We have no non-invasive rule that safely spares a child a screening endoscopy, and the evidence for preventing a first variceal bleed in children rests on one sclerotherapy trial and no beta-blocker trial. Universal screening for infant cholestasis has not been implemented in most of the United States and Europe, and late Kasai operations still happen. And a variant of uncertain significance takes on average about two years to be resolved, which shared paediatric variant databases could shorten.

Part seven. What to do differently in clinic.

So what changes tomorrow? Write the question on the request form: is this test meant to diagnose, to stage, to predict, or to decide? Fractionate every bilirubin after two weeks of age, and look at the stool yourself rather than relying on a description. Give vitamin K before you believe an INR, and stop treating INR numbers before procedures. Ask the laboratory for the core length and the number of portal tracts, and give the pathologist the age, the GGT and the question. Read liver stiffness as a trend from the child's own baseline, measured after inflammation has settled. Never quote an adult serum score or stiffness cut-off to a parent as if it applied to their child. Send the trio, phase every unresolved variant, and never let a pending result delay treatment. And book an endoscopy or a cholangiogram only when its result will change the plan, sending the smallest infants to the centre that does these procedures every month.

Questions to test yourself.

Question one. A twenty-one-day-old infant has a total bilirubin of three and a direct bilirubin of one point two milligrams per decilitre. Is this abnormal? Answer: yes. A direct bilirubin above one milligram per decilitre is pathological at any total, and the infant needs evaluation.

Question two. A HIDA scan shows activity in the bowel at four hours. What does that exclude? Answer: biliary atresia, because a non-draining scan has a sensitivity of about ninety-nine percent, so a draining scan makes atresia very unlikely.

Question three. How many portal tracts are needed to make a diagnosis, and how many to stage fibrosis? Answer: about six to diagnose, at least eleven to stage according to AASLD, and ten for biliary atresia according to ESPGHAN.

Question four. A cholestatic infant has an INR of two. What is the first step? Answer: parenteral vitamin K, with a repeat INR the next day, and factor five if the INR is still prolonged.

Question five. A well boy has an isolated AST three times the upper limit of normal. Which two tests come first? Answer: creatine kinase, and then polyethylene glycol precipitation for macro-AST.

Question six. A child has a transient elastography reading of sixteen kilopascals during acute hepatitis. Does she have cirrhosis? Answer: not necessarily. Repeat the measurement in remission on the same machine.

Question seven. What is the approximate upper limit of normal transient elastography in a healthy child? Answer: about five and a half kilopascals, and higher with sedation, the small probe and liver fat.

Question eight. A four-month-old with cholestasis has an APRI of one point two. What does it mean? Answer: it cannot be interpreted. There is no validated infant cut-off, and infant GGT and AST norms differ from those of adults.

Question nine. A gene panel shows one pathogenic variant and one variant of uncertain significance in ABCB11. What are the next two steps? Answer: phase the variants by testing the parents, and stain the liver tissue for the bile salt export pump. Treatment should not wait for either.

Question ten. A fourteen-month-old with biliary atresia has falling platelets. When should she have an endoscopy? Answer: now, because three quarters of first variceal bleeds happen before two and a half years of age.

End of chapter.